Voice Assistance for Multimedia Control

Hardware-Based Voice Control System using Raspberry Pi / ESP32
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1. Software Requirements Specification

1. Introduction

1.1 Purpose

This document specifies the software requirements for a Voice Assistance system for Multimedia Control. The system allows a user to control local media playback using spoken voice commands, processed on embedded hardware (Raspberry Pi or ESP32), instead of physical buttons or a mouse/keyboard. This SRS is intended for the project development team, the course evaluator, and any future maintainer of the system.

1.2 Scope

The system, referred to hereafter as the “Voice Media Controller”, will:

- Continuously listen for a wake word / voice command through a connected microphone.
- Recognize a fixed set of multimedia control commands: Play, Pause, Resume, Next, Previous, Volume Up, and Volume Down.
- Execute the recognized command on a connected media player (e.g. a local audio/video player running on the same device or a paired system).
- Run on low-cost embedded hardware (Raspberry Pi / ESP32) using Python and a speech/audio processing library (e.g. MediaPipe, Speech Recognition, or equivalent).

The system will NOT perform general-purpose conversational AI, internet search, smart-home control, or support commands outside the defined multimedia control set in this version.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification

FR	Functional Requirement
NFR	Non-Functional Requirement
Pi	Raspberry Pi single-board computer
ESP	ESP32 / ESP8266 microcontroller board
ASR	Automatic Speech Recognition
Wake Word	A specific trigger word/phrase that activates command listening
Media Player	The audio/video playback application controlled by the system

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Raspberry Pi Foundation, Raspberry Pi Documentation, www.raspberrypi.com/documentation
- Espressif Systems, ESP32 Technical Reference Manual.
- Project handout: SE Mini-Project Deliverables Part-1.

1.5 Overview

The remainder of this document is organized as follows: Section 2 gives an overall description of the product, its perspective, functions, users and constraints. Section 3 lists the specific functional and non-functional requirements along with the use case diagram. Section 4 details the security objectives and requirements for the system.

2. Overall Description

2.1 Product Perspective

The Voice Media Controller is a standalone embedded add-on device. It sits between the user and an existing media player, listening for audio input through a microphone and issuing control signals to the media player through a software or serial interface. It is not a replacement for the media player itself, only a hands-free control layer on top of it.

2.2 Product Functions

At a high level, the system:

- Captures live audio from a microphone connected to the Pi/ESP hardware.
- Converts speech to a recognized command using an ASR module.
- Maps the recognized command to one of: Play, Pause, Resume, Next, Previous, Volume Up, Volume Down.
- Sends the corresponding control signal to the media player.
- Provides basic feedback (e.g. an LED or console log) confirming the command executed.

2.3 User Characteristics

The primary user is a single end user operating the media player hands-free, for example while cooking, exercising, or working. No technical expertise is required to use the system. Only the developer/maintainer requires technical knowledge of Python, embedded hardware, and audio processing.

2.4 Constraints

- Limited to the fixed command vocabulary defined in Section 1.2; free-form conversation is out of scope.
- Hardware constraints: processing power and memory available on Raspberry Pi / ESP32 limit the choice of ASR model (lightweight/offline models preferred over large cloud models).
- Requires a working microphone and, if used, a stable network connection for any cloud-based recognition fallback.
- Implementation of language restricted to Python (with C/C++ for ESP-side firmware where required).

2.5 Assumptions and Dependencies

- The target media player exposes a controllable interface (CLI, API, or keypress simulation) that the system can call.
- The operating environment has low to moderate background noise.
- The Raspberry Pi / ESP32 has the required libraries (e.g. MediaPipe / SpeechRecognition) pre-installed or installable.

- A single user issues commands at a time; multi-user concurrent commands are not handled in this version.

3. Specific Requirements

3.1 Functional Requirements

Functional requirements are written to be clear, unambiguous, measurable, and independently testable, in line with IEEE 830 guidelines.

ID	Requirement	Description / Acceptance Criteria
FR-1	Wake / Listen	The system shall continuously monitor the microphone input and begin command processing within 1 second of detecting the wake word.
FR-2	Play Command	The system shall start media playback within 2 seconds of recognizing the spoken command “Play”.
FR-3	Pause Command	The system shall pause the currently playing media within 2 seconds of recognizing the spoken command “Pause”.
FR-4	Resume Command	The system shall resume previously paused media within 2 seconds of recognizing the spoken command “Resume”.
FR-5	Next Track Command	The system shall skip to the next media item within 2 seconds of recognizing the spoken command “Next”.
FR-6	Previous Track Command	The system shall return to the previous media item within 2 seconds of recognizing the spoken command “Previous”.
FR-7	Volume Up Command	The system shall increase playback volume by a fixed step (10%) within 1 second of recognizing the spoken command “Volume Up”.
FR-8	Volume Down Command	The system shall decrease playback volume by a fixed step (10%) within 1 second of recognizing the spoken command “Volume Down”.

FR-9	Command Rejection	The system shall ignore and discard any audio input that does not match a defined command in the vocabulary, without executing an action.
FR-10	Feedback	The system shall provide a visible or audible acknowledgement (e.g. LED blink or log message) each time a command is successfully executed.

3.2 Non-Functional Requirements

ID	Requirement	Description / Acceptance Criteria
NFR-1	Performance	The system shall recognize and execute a valid voice command within 2 seconds under normal (low) background noise, measured from end of utterance to action execution.
NFR-2	Reliability	The system shall correctly recognize valid commands with at least 90% accuracy across 50 test utterances in a quiet room.
NFR-3	Availability	The system shall remain in a listening-ready state for at least 8 continuous hours of operation without requiring a restart.
NFR-4	Usability	A first-time user shall be able to issue a working command without reading a manual, using only the 7 defined command words.
NFR-5	Portability	The core recognition and control software shall run on both Raspberry Pi (Linux/Python) and ESP32 (with minimal, documented code changes) without a redesign.
NFR-6	Maintainability	Each voice command shall be defined as a separate, independently testable function/module so new commands can be added without modifying existing command logic.

3.3 Use Case Diagram

Figure 1 shows the primary actor (User) and the system's use cases. All direct commands include the "Recognize Voice Command" use case, which in turn includes "Authenticate Command Source" to satisfy the security requirements in Section 4.

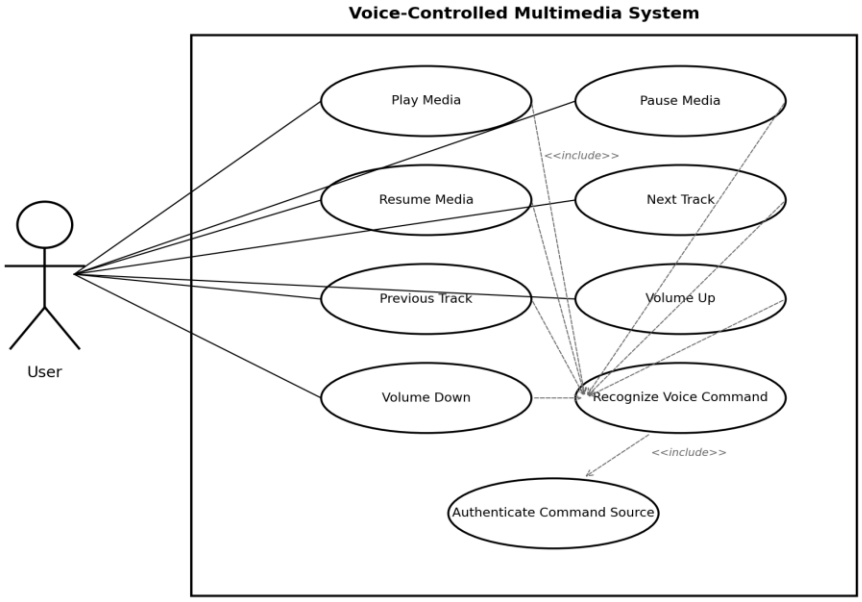


Figure 1: UML Use Case Diagram — Voice-Controlled Multimedia System

4. Security Requirements

4.1 Security Objectives

ID	Objective	Description

SO-1	Command Source Integrity	Ensure that multimedia control commands are only accepted from the intended, authorized microphone/user, preventing accidental or malicious triggering from other audio sources (e.g. a TV or radio in the background).
SO-2	Communication Confidentiality	Protect any command or audio data transmitted between the microphone module, the processing unit (Pi/ESP), and the media player from interception or tampering.

4.2 Security Requirements

ID	Requirement	Description
SR-1	Wake-Word Gating	The system shall require detection of a specific wake word before accepting any control command, reducing the chance of unintended activation by background audio.
SR-2	Local Processing	The system shall process voice commands locally on the Pi/ESP device wherever feasible, avoiding transmission of raw audio to external/cloud services unless explicitly required.